

programming language called Microbe. It is basically a compiler for the PIC microcontrollers. Due to its similarity to Basic language, it is also called MicroBasic. It acts as a companion program to KTechlab. PIC16F84, PIC16F628, PIC16F627 and PIC16F877 are the microcontrollers' Microbe supports.

Assembler and disassembler. KTechlab can integrate both an assembler and a disassembler via gpsasm and gpdasm.

What users think about this tool

Every designer reviews a tool before using it, in order to check if it would be of any use to him. No one can get a better review than from the users and the developers themselves. Following are the remarks from sourceforge.net, kde-apps.org and edaboard.com communities:

1. "This is terrific program. Comparable to MultiSim by National Instruments. I use it on Kubuntu 9.04 AMD64. I plan on installing it on my laptop with Kubuntu 10.4 to see how it works. Thanks for a great program. Keep up the good work."

2. "Great application that's a pleasure to use! It compiled and ran perfectly on SUSE 10.2. Now if we can just get PCB design support, that would be fantastic!"

How to install KTechlab

Following is the way to compile and then install KTechlab on Linux:

1. Use a terminal and navigate to the top-level source directory
2. Run the shell script:

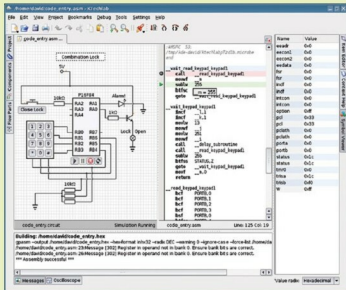


Fig. 3: Compilation and simulation side by side

simple-setup.sh

This command compiles KTechlab into the directory *simple-build*, installs it into the directory *simple-install* and sets up the user-specific settings (sysconfig, mime database) for the current user.

```
sh simple-setup.sh
```

3. Launch KTechlab by running the script *simple-launch.sh*

```
sh simple-launch.sh
```

KTechLab should start running at this point.

Installing KTechlab on Ubuntu. We can also install this tool on the latest version of Ubuntu. However, we need to use the Debian version:

```
http://launchpadlibrarian.net/50881922/ktechlab_0.3.7-9ubuntu1_1386.deb
```

To download this package we have to install wget:

```
apt-get install wget
```

Then we can download *ktechlab_0.3.7-9ubuntu1_1386.deb* using the command:

```
wget http://old.releases.ubuntu.com/ubuntu/pool/universe/k/ktechlab/ktechlab_0.3.7-9ubuntu1_1386.deb
```

Enter the following command to install *ktechlab_0.3.7-9ubuntu1_1386.deb*:

```
dpkg -i ktechlab_0.3.7-9ubuntu1_1386.deb
```

A claim of dependence will appear as *libpopt0*, but we can ignore it.

After the installation, exit the

System requirements

KTechlab does not have any special system requirements. It is compatible with operating systems like Linux, BSD and Unix, though it has following dependencies for better usage:

- Qt4-dev
- cmake
- glib-dev

Then there are a few optional dependencies too:

- KDE4 libraries-dev
- KDevPlatform libraries-dev

Note. Only unit tests will get built without the above-listed components, and these are not useful for end users.

Users should install 'GPSim-dev' and 'Eigen2-dev' (included in the DVD) to make full use of this tool.

➤ GPSim-dev; for microcontroller support

➤ Eigen2-dev; for unit tests; generally it is not needed for builds targeted to end-users, but it is useful for development

Competitive tools

- MPLAB IDE
- PIC Simulator IDE
- Pikelab

Then there are a few enterprise versions of the IDEs for PIC microcontroller:

- IAR Workbench IDE
- CCS' IDE
- Mikroelektronika's IDE

Ubuntu Lucid:

```
exit
```

Now, to run KTechlab enter the command:

```
schroot p KTechlab
```

Allowing multiple users on same build. An already compiled and installed version of KTechlab allows multiple-user access. In order to do this, run the script:

```
ktechlab-user-setup.sh
```

from the *simple-install/bin/* directory:

```
sh simple/install/bin/ktechlab-user-setup.sh
```

Then you can launch KTechlab as usual:

```
sh simple-launch.sh
```

The author is a technical journalist at EFY. The article is based on the KTechlab documents available at sourceforge.net, space-kerala.org and github.com